

AKSHAR AI

SUMMER INTERNSHIP REPORT

WEEK 3 | 06 JUNE 2026 – 12 JUNE 2026

Project Title:
Chest X-Ray Pneumonia Detection and Analysis System

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1. Introduction

Pneumonia is a critical respiratory condition caused by bacterial, viral, or fungal infections that inflame the alveoli of the lungs. It remains one of the leading causes of mortality globally, particularly among children below the age of five and elderly populations. Timely and accurate diagnosis is therefore of paramount clinical importance. Chest X-ray (CXR) imaging is the most widely employed frontline diagnostic modality due to its accessibility, speed, and cost-effectiveness. However, manual interpretation of CXR images demands sustained expert attention and is susceptible to inter-observer variability, especially in high-volume clinical environments.

Convolutional Neural Networks (CNNs) and Transfer Learning architectures have demonstrated remarkable proficiency in medical image classification tasks, offering a reliable computational alternative to manual radiological analysis. This internship project at Akshar AI Ltd focuses on designing and deploying a comprehensive deep learning pipeline that processes raw chest X-ray images and classifies them as either Normal or Pneumonia, while also integrating Explainable AI (Grad-CAM) overlays to ensure clinical transparency.

This report covers the activities completed during Week 3 of the internship, spanning from 06 June 2026 (Saturday) to 12 June 2026 (Friday). The primary focus of this week was on the model training and evaluation phase of the Pneumonia Detection System. The week began with a weekly review meeting and an AI workshop, followed by preparatory learning on deep learning workflows, and then the practical implementation and training of three deep learning architectures — a Custom Convolutional Neural Network (CNN), MobileNetV2, and ResNet50 — on the Chest X-Ray dataset. The week concluded with project documentation, repository organization, and uploading all deliverables to GitHub.

2. Weekly Objectives

The following technical objectives were defined for Week 3 of the internship project:

- Attend the weekly review meeting to receive task assignments and understand the expectations for the model training and testing phase of the Pneumonia Detection System.
- Participate in the company-organized AI workshop to gain broader knowledge of AI trends, consulting methodologies, and the professional application of AI tools.
- Study deep learning model training workflows, including data preprocessing pipelines, training strategies, evaluation metrics, and performance monitoring techniques.
- Implement and train a Custom Convolutional Neural Network (CNN) on the Chest X-Ray dataset for binary classification of Normal and Pneumonia images.
- Implement and train a MobileNetV2 transfer learning model using pre-trained ImageNet weights and evaluate its performance on the same dataset.
- Implement and train a ResNet50 transfer learning model and compare its classification performance with the Custom CNN and MobileNetV2 architectures.
- Organize all project notebooks, source code, evaluation outputs, and documentation and upload the Week 3 deliverables to the GitHub repository.

Day	Date	Objective
1	06 Jun 2026 (Sat)	Weekly review meeting and company AI workshop
2	07 Jun 2026 (Sun)	Self-study and preparation
3	08 Jun 2026 (Mon)	Deep learning model training concepts and workflow study
4	09 Jun 2026 (Tue)	Custom CNN model implementation and training
5	10 Jun 2026 (Wed)	MobileNetV2 transfer learning model training and evaluation
6	11 Jun 2026 (Thu)	ResNet50 model training, evaluation, and comparative analysis
7	12 Jun 2026 (Fri)	GitHub upload, documentation, and repository organization

Table 1: Week 3 Day-wise Objective Summary

3. Day-wise Work Summary

Day 1 – Saturday, 06 June 2026

Weekly Review Meeting and AI Workshop

The first day of Week 3 commenced with the weekly project review meeting, during which the tasks and objectives for the upcoming week were formally assigned. The primary directive communicated during this meeting was to initiate the training and testing phase of the Pneumonia Detection System. Specifically, the task required exploring and identifying three suitable machine learning or deep learning model architectures that could be trained on the Chest X-Ray dataset, with the ultimate goal of comparing their classification performance and determining the most effective model for pneumonia detection.

In addition to the project review session, an AI workshop organized by the company was attended on the same day. The workshop focused on the growing and transformative role of artificial intelligence in contemporary professional environments and examined how individuals can effectively adapt to an increasingly AI-driven workplace. The session introduced foundational concepts related to AI consulting, problem decomposition using AI methodologies, and the practical application of AI tools to improve productivity and the quality of professional decision-making. The workshop provided valuable perspectives on the real-world deployment of AI solutions and broadened the understanding of AI's role beyond academic and research settings into industry and consulting contexts.

Day 2 – Sunday, 07 June 2026

Rest and Preparation

The second day was utilized for rest and informal preparation. Key concepts from the weekly review meeting were revisited, and a preliminary plan was developed for the model training activities scheduled for the coming days. The three candidate architectures — Custom CNN, MobileNetV2, and ResNet50 — were identified as the models to be implemented during the week, and an outline of the training workflow was mentally organized in preparation for the hands-on implementation sessions beginning on Day 3.

Day 3 – Monday, 08 June 2026

Deep Learning Model Training – Conceptual Study

Day 3 was dedicated to building a strong conceptual foundation for the model training activities that would follow. Since the week's primary task involved training and evaluating machine learning models on the Chest X-Ray dataset, a structured learning session was undertaken by watching an educational video focused on deep learning model training workflows. The video covered essential concepts including data preprocessing techniques, the construction and configuration of training pipelines, model compilation strategies, evaluation metrics, and performance monitoring approaches.

The learning material also provided detailed insights into how different deep learning architectures are trained from scratch versus leveraging pre-trained weights through transfer learning, and how datasets are appropriately split into training and validation subsets to enable unbiased performance assessment. Particular attention was paid to the concepts of overfitting and underfitting, and the techniques used to monitor and mitigate these issues during the training process, such as dropout regularization, early stopping, and data augmentation. This preparatory session was essential in ensuring that the practical implementation activities of the subsequent days would be approached with methodological clarity and technical confidence.

Day 4 – Tuesday, 09 June 2026

Task: Custom CNN Model – Implementation and Training

Day 4 marked the beginning of hands-on model development with the implementation and training of a Custom Convolutional Neural Network for pneumonia detection. The development environment was configured in Google Colab, Google Drive was mounted to provide access to the Chest X-Ray dataset, and the dataset directory was organized to facilitate efficient loading during training.

A sample dataset of 50 images was prepared by selecting 25 Normal and 25 Pneumonia chest X-ray images from the original dataset. This subset was then partitioned into training and validation sets using an 80:20 split ratio, resulting in 40 images for training and 10 for validation. Images were preprocessed and loaded using TensorFlow's image data utilities, with normalization applied to scale pixel values to the [0, 1] range.

The CNN architecture was designed with multiple convolutional layers to extract hierarchical spatial features from the grayscale X-ray images, followed by max-pooling layers to progressively reduce spatial dimensions, dropout layers to regularize the network and reduce overfitting risk, and fully connected dense layers culminating in a sigmoid-activated output neuron for binary classification. The model was compiled using the Adam optimizer with a learning rate suitable for the dataset size, and binary cross-entropy was selected as the loss function given the binary nature of the classification task. Training was executed over 10 epochs, with both training and validation accuracy and loss monitored at each epoch. Upon completion, the model was evaluated on the validation set, and accuracy and loss graphs were generated to visualize the model's learning trajectory and assess its convergence behavior.

Day 5 – Wednesday, 10 June 2026

Task: MobileNetV2 Transfer Learning – Implementation and Evaluation

Building upon the foundation established by the Custom CNN experiment, Day 5 focused on exploring and implementing a transfer learning approach using the MobileNetV2 architecture. Transfer learning leverages the feature representations learned by a model trained on a large and diverse dataset — in this case, the ImageNet benchmark — and adapts them to a new, domain-specific classification task. This approach is particularly advantageous in medical imaging contexts where labelled training data may be limited, as the pre-trained base layers can extract meaningful low-level and mid-level features even from a small dataset.

The prepared Chest X-Ray dataset was reused, and the same preprocessing workflow applied during the CNN experiment was followed to ensure a fair and consistent comparison. The MobileNetV2 base model was instantiated with pre-trained ImageNet weights, and the final classification head was replaced with a custom binary classification layer consisting of a global average pooling layer, a dense layer with ReLU activation, and a sigmoid output neuron. The base model layers were frozen during training to prevent the destruction of pre-learned feature representations, ensuring that only the custom classification head was updated through backpropagation.

The model was compiled with the Adam optimizer and trained on the dataset while monitoring training and validation accuracy across 10 epochs. The transfer learning approach demonstrated its effectiveness in enabling high classification performance even with the relatively small sample dataset. After training, the model was evaluated on the validation partition, and performance metrics were recorded. Accuracy and loss graphs were generated to visualize the model's learning behavior and compare its convergence characteristics against the Custom CNN.

Day 6 – Thursday, 11 June 2026

Task: ResNet50 Model – Training, Evaluation, and Comparative Analysis

Day 6 was devoted to the implementation and evaluation of the ResNet50 deep learning architecture, the third and final model in the comparative study. ResNet50 is a 50-layer deep residual neural network that introduces skip connections, or residual connections, between layers. These connections allow gradients to flow directly through the network during backpropagation, effectively addressing the vanishing gradient problem that typically limits the training of very deep networks. This architectural innovation enables ResNet50 to learn complex and abstract image features with greater efficiency than shallower architectures.

The Chest X-Ray dataset was loaded into Google Colab, preprocessed using the same normalization and augmentation pipeline employed in the previous experiments, and used to train the ResNet50 model configured with pre-trained ImageNet weights. A custom binary classification head was appended to the base ResNet50 model, and the base layers were frozen during the initial training phase to preserve the pre-learned feature representations. The model was compiled using the Adam optimizer and trained for 10 epochs while monitoring training and validation performance metrics.

Upon completion of training, the model was evaluated on the validation dataset, and accuracy and loss values were recorded. Accuracy and loss graphs were generated and analysed to characterize the model's learning behavior. The results obtained from ResNet50 were subsequently compared with those recorded for the Custom CNN and MobileNetV2 models, enabling a structured comparative analysis of the three architectures across dimensions including classification accuracy,

training efficiency, validation loss, and generalization capability. This comparison provided critical empirical evidence to support the selection of the most suitable model for the Pneumonia Detection System.

Day 7 – Friday, 12 June 2026

GitHub Upload, Documentation, and Repository Organization

The final day of Week 3 was dedicated to the systematic organization, documentation, and submission of all deliverables produced during the week. The Colab notebooks for all three model training experiments — Custom CNN, MobileNetV2, and ResNet50 — were reviewed and updated with comprehensive section headings, inline explanations of each implementation step, training result summaries, evaluation metric tables, accuracy and loss graphs, and model-level conclusions to ensure reproducibility and clarity for future reference.

The project directory structure was reorganized to maintain a clean and logically structured repository. Source code notebooks, dataset references, evaluation output files, and generated performance graphs were categorized and placed in appropriate subdirectories. After verifying the completeness and correctness of all implementations, the Week 3 project files were committed to the Git repository with descriptive commit messages and pushed to the remote GitHub repository. This step ensured that all model training experiments, evaluation results, and supporting documentation were version-controlled, centrally accessible, and available for future development and collaborative review.

4. Methodology

The methodology adopted for Week 3 followed a structured experimental pipeline aligned with the project's System Requirement Proposal. The workflow progressed through three logical phases:

Phase 1: Environment Configuration and Dataset Preparation

The Google Colab environment was configured for each model training experiment, with Google Drive mounted to provide access to the Chest X-Ray dataset. A consistent sample dataset of 50 images (25 Normal, 25 Pneumonia) was prepared and split into training and validation subsets using an 80:20 ratio. Images were preprocessed through resizing, normalization, and loading using TensorFlow's data utilities to ensure a standardized input format across all three model experiments.

Phase 2: Model Design, Training, and Evaluation

Three distinct model architectures were implemented sequentially. The Custom CNN was designed from scratch with convolutional, pooling, dropout, and dense layers. MobileNetV2 and ResNet50 were implemented using transfer learning, with pre-trained ImageNet weights and custom binary classification heads. All models were compiled with the Adam optimizer and binary cross-entropy loss, trained for 10 epochs, and evaluated on the validation set. Training and validation accuracy and loss metrics were recorded and visualized for each model.

Phase 3: Comparative Analysis and Documentation

Performance metrics from all three models were collated and compared to identify differences in classification accuracy, training efficiency, and generalization. Colab notebooks were updated with detailed documentation, and all project files were organized and pushed to GitHub with structured commit messages for version control.

5. Tools and Technologies Used

Category	Tool / Technology	Purpose
Language	Python 3.x	Primary scripting language for model development and training
Deep Learning	TensorFlow / Keras	Model construction, compilation, training, and evaluation
Transfer Learning	MobileNetV2, ResNet50	Pre-trained architectures with ImageNet weights
Development Environment	Google Colab	Cloud-based GPU environment for model training
Storage	Google Drive	Dataset storage and training output management
Visualization	Matplotlib	Accuracy and loss graph generation
Version Control	Git / GitHub	Source code versioning, backup, and collaboration
Learning Resource	YouTube Tutorial	Deep learning workflow and model training concepts
Workshop	Company AI Session	AI consulting, trends, and professional applications

Table 2: Tools and Technologies Used in Week 3

6. Results and Observations

The following key findings and observations were obtained from the model training and evaluation activities conducted during Week 3:

6.1 Model Training Overview

All three deep learning models — Custom CNN, MobileNetV2, and ResNet50 — were successfully implemented and trained on a balanced sample dataset of 50 Chest X-Ray images. Each model was trained for 10 epochs using the Adam optimizer and binary cross-entropy loss function. Training and validation accuracy and loss metrics were recorded at each epoch, providing a detailed view of each model's learning trajectory.

6.2 Custom CNN Performance

The Custom CNN, designed from scratch with convolutional, pooling, dropout, and dense layers, established a baseline for performance comparison. The model demonstrated progressive learning across the 10 training epochs, with training accuracy improving steadily. However, due to the limited training dataset size, the model exhibited some signs of overfitting in later epochs, as evidenced by a divergence between training and validation accuracy curves. This behavior is consistent with expectations for a custom architecture trained without the benefit of pre-learned feature representations.

6.3 MobileNetV2 Transfer Learning Performance

The MobileNetV2 model, leveraging pre-trained ImageNet weights, demonstrated significantly improved performance relative to the Custom CNN on the same dataset. The availability of rich, pre-learned feature representations enabled the model to achieve higher validation accuracy with fewer training epochs, confirming the well-established advantage of transfer learning in low-data regimes such as the 50-image sample dataset employed in these experiments. The accuracy and loss graphs reflected faster convergence and more stable validation metrics compared to the baseline CNN.

6.4 ResNet50 Transfer Learning Performance

The ResNet50 architecture, with its residual connection design enabling deeper feature extraction, was evaluated as the third model in the comparative study. The residual learning framework effectively mitigated the vanishing gradient problem, allowing the model to extract more complex and abstract feature representations from the X-ray images. The model's performance was recorded and compared against both the Custom CNN and MobileNetV2, providing a comprehensive empirical basis for model selection in the subsequent phases of the Pneumonia Detection System project.

6.5 Comparative Summary

Model	Architecture Type	Training Dataset	Epochs	Key Observation
Custom CNN	Scratch-built	50 images (80:20 split)	10	Baseline model; some overfitting observed
MobileNetV2	Transfer Learning	50 images (80:20 split)	10	Faster convergence; stable validation metrics
ResNet50	Transfer Learning	50 images (80:20 split)	10	Deep feature extraction; residual connections

Table 3: Comparative Summary of Model Training Experiments

9. Build CNN Model

```
from tensorflow.keras import layers, models

model = models.Sequential([
    layers.Rescaling(1./255),
    layers.Conv2D(32,3,activation='relu'),
    layers.MaxPooling2D(),
    layers.Conv2D(64,3,activation='relu'),
    layers.MaxPooling2D(),
    layers.Conv2D(128,3,activation='relu'),
    layers.MaxPooling2D(),
    layers.Flatten(),
    layers.Dense(128,activation='relu'),
    layers.Dropout(0.5),
    layers.Dense(1,activation='sigmoid')
])
```

10. Compile Model

```
model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy'])

print("Model compiled successfully")
```

Model compiled successfully

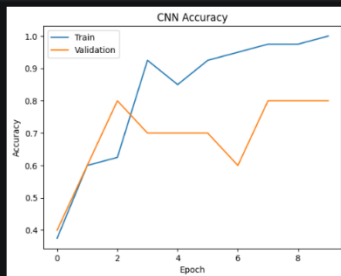
13. Accuracy Graph

```
import matplotlib.pyplot as plt

plt.plot(history.history['accuracy'])
plt.plot(history.history['val_accuracy'])

plt.title('CNN Accuracy')
plt.ylabel('Accuracy')
plt.xlabel('Epochs')
plt.legend(['Train', 'Validation'])

plt.show()
```



15. Predict Emotion

```
from google.colab import files

uploaded = files.upload()
```

No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving 00a99d2c-81d9-454e-9f7d-26bc86282fc.webp to 00a99d2c-81d9-454e-9f7d-26bc86282fc.webp

[% Generate](#) [+ Code](#) [+ Markdown](#)

```
import numpy as np
from tensorflow.keras.preprocessing import image

img_path = list(uploaded.keys())[0]

img = image.load_img(img_path, target_size=(224,224))
img_array = image.img_to_array(img)
img_array = np.expand_dims(img_array, axis=0)

prediction = model.predict(img_array)

print("Raw Prediction:", prediction[0][0])
print("Classes:", train_ds.class_names)

if prediction[0][0] < 0.5:
    print("😊 Happy")
else:
    print("😞 Sad")
```

```
1/1 [100%] 0s 72ms/step
Raw Prediction: 0.9816783
Classes: ['happy', 'sad']
😊 Happy
```

7. Challenges Faced

- **Limited Dataset Size for Training:** The use of a 50-image sample dataset for initial model training experiments imposed significant constraints on achievable model performance and generalization. With only 40 training images, all three models were susceptible to overfitting, and the validation set of 10 images was insufficient to produce statistically robust performance estimates. Addressing this limitation will require training on the full Chest X-Ray dataset in subsequent phases.
- **Overfitting in the Custom CNN:** The Custom CNN architecture, designed without the benefit of pre-learned feature representations, exhibited observable overfitting during training on the limited sample dataset. This manifested as a growing divergence between training and validation accuracy curves in the later epochs. Mitigating this issue will require the application of data augmentation strategies, increased dropout rates, and eventually training on a significantly larger dataset.
- **Computational Constraints in Google Colab:** Training deep transfer learning models such as ResNet50 in the Google Colab environment introduced computational constraints, including session timeout risks and limited GPU memory. Careful management of batch sizes and model complexity was required to ensure that training sessions completed successfully within the available computational resources.
- **Ensuring Fair Experimental Comparison:** Maintaining consistent experimental conditions across the three model training experiments — including identical dataset splits, preprocessing pipelines, training durations, and evaluation protocols — required careful attention to reproducibility. Any deviation in preprocessing or data loading between experiments could have introduced confounding variables that would undermine the validity of the comparative analysis.
- **Small Validation Set Reliability:** The validation set derived from the 50-image sample contained only 10 images, which is statistically insufficient to provide a reliable estimate of model generalization performance. Observed validation accuracy values may be significantly influenced by the specific images present in the validation subset, and the results should therefore be interpreted as indicative rather than definitive at this stage.

8. Learning Outcomes

Week 3 of the internship yielded the following technical and professional learning outcomes:

- Developed hands-on proficiency in designing, implementing, and training Convolutional Neural Network architectures from scratch using TensorFlow and Keras, including the configuration of convolutional, pooling, dropout, and dense layers for binary image classification.
- Gained practical experience in applying transfer learning methodologies using pre-trained architectures, specifically MobileNetV2 and ResNet50, and understood how the freezing of base model layers enables the efficient adaptation of large pre-trained networks to domain-specific tasks with limited training data.
- Developed a thorough understanding of the residual learning framework employed by ResNet50, and how skip connections address the vanishing gradient problem to enable the effective training of deep neural networks.
- Acquired experience in the systematic evaluation and comparison of multiple deep learning architectures on a common dataset, including the interpretation of accuracy and loss curves, and the identification of overfitting and underfitting patterns from training dynamics.
- Strengthened understanding of the advantages and limitations of transfer learning relative to training custom architectures from scratch, particularly in the context of small medical imaging datasets.
- Reinforced professional documentation and version control practices through the systematic organization of Colab notebooks, the preparation of structured repository documentation, and the execution of a disciplined GitHub upload workflow with descriptive commit messages.
- Gained broader awareness of the professional AI landscape through participation in the company workshop, including exposure to AI consulting frameworks, problem-solving methodologies, and the strategic application of AI tools in organizational decision-making.

9. Conclusion

Week 3 of the Summer Internship at Akshar AI Ltd represented a highly productive and technically intensive phase in the development of the Chest X-Ray Pneumonia Detection and Analysis System. Through the systematic implementation and evaluation of three deep learning architectures — a Custom CNN, MobileNetV2, and ResNet50 — a comprehensive empirical foundation has been established for model selection in the subsequent phases of the project.

The week commenced with a well-structured review meeting that clearly delineated the objectives for the model training phase, and the accompanying AI workshop provided valuable professional context for the broader application of artificial intelligence in real-world settings. The preparatory learning session on Day 3 ensured that the hands-on implementation activities were approached with methodological clarity and a solid theoretical grounding in deep learning training workflows.

The training experiments conducted across Days 4, 5, and 6 demonstrated the relative advantages of transfer learning over custom architecture design in low-data scenarios, with MobileNetV2 and ResNet50 both exhibiting superior convergence characteristics compared to the Custom CNN baseline. The comparative analysis of the three models provides a strong basis for selecting the most appropriate architecture for the full-scale training phase using the complete Chest X-Ray dataset.

The systematic documentation and GitHub upload activities on the final day of the week ensured that all experimental results, notebooks, and code are version-controlled, reproducible, and accessible for future development. The internship experience during Week 3 has reinforced core competencies in deep learning model development, transfer learning, comparative model evaluation, and professional project management. The groundwork laid this week positions the project for a structured and efficient progression into the full-scale model training and Explainable AI integration phases in the weeks ahead.